MIL-P-16208D

29 September 1969
SUPERSEDING
MIL-P-16208C
30 March 1965

MILITARY SPECIFICATION

PUNCHES AND DIES, GROMMET INSERTING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers grommet inserting punches and dies.

* 1.2 Classification.- The punches and dies shall be of the following types, classes and sizes, as specified (see 6.2).

Type I - Eyelet grommet inserting
Type II - Spur grommet inserting

Class-1 with integral dies
Class-2 with inserted dies

Sizes (see paragraph 3.4 and applicable figures)

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-H-15424 - Hand Tools, Packaging of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 5120

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18-65 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), before production is commenced, the supplier shall submit a preproduction sample to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

* 3.2 Standard product.- The punches and dies covered by this specification shall be the manufacturer's current commercial product except for any changes necessary to comply with this specification.

* 3.3 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for punches and dies, provided the completed item complies with all provisions of this specification.

* 3.4 Design and construction.- The type I, and type II, class 1 and 2 punches and dies and die holder (when applicable) shall be of same designated size, as specified (see 6.2). The finished surfaces of punches and dies shall be concentric within 0.002 inch. Punches, dies, and die holders shall conform to the design and construction specified herein, and as shown on the applicable figures as follows:

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Type I punch and die - Figure 1 and figure 2

Type II, class 1, punch and die - Figure 3 and figure 4

Type II, class 2, punch, die and die holder - Figure 5 and figure 6

* 3.4.1 Punches.- Type I, and type II, class 1 and 2 punches shall be formed of steel. The handle portion of the punch may be made of malleable iron. The punch end shall have a hardness of Rockwell C 42-50 inclusive, or may be casehardened not less than 0.010 inch deep and have a superficial hardness of Rockwell 15N, 81.5, - 85.5, inclusive, when tested as specified in 4.4.1.

* 3.4.2 Die and die holders.- Type I and type II, class 1 and 2 dies shall be made of steel. Die Holders for type II, class 2 dies shall be made of cast iron with an inserted hardened die. Dies shall have a hardness of Rockwell C 42-50 inclusive, or type I, and type II, class 1 dies may be casehardened not less than 0.010 inch deep and have a superficial hardness of Rockwell 15N, 81.5, - 85.5 inclusive, when tested as specified in 4.4.1.

* 3.5 Finish.- Working surfaces of punches and dies shall be polished to a surface roughness value of 63 microinches or better. Other surfaces shall be in accordance with the manufacturer's current commercial practice.

* 3.6 Marking.- The punch and die shall be marked with "eyelet" or "spur" and the size designation, as applicable. In addition, the manufacturer shall permanently and legibly indent his name, or trademark of such known character as to be easily identifiable to the manufacturer on the punch and die.

* 3.7 Workmanship.- Punches and dies shall be free from defects such as cracks, pits, inclusion of foreign matter, burrs, rough or sharp edges. Working surfaces shall be free of tool marks.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects in table I, specified dimensions and tested as specified in 4.4.1. The presence of any visual defect, any dimension not within the specified requirements, or failure to pass the test shall be cause for rejection of the preproduction sample.

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* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification, or applicable purchase document.

* 4.3.2 End item inspection.- A lot shall consist of all punches and dies of one type, class, and size offered for inspection at one time. The sample unit for this inspection shall be one complete punch and die.

* 4.3.2.1 Visual examination.- Examination shall be made of the punch and dies for defects set forth in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Visual examination of defects

Examine	Defect	Classification Major	Minor
Finish	Working surfaces not polished to 63 microinches or better	X	
Construction and workmanship	Crack, pit, burr, rough or sharp edge	X	
	Evidence of imbedded foreign matter		X
	Tool mark on working surface	X	
Marking	Missing, illegible, not as specified		X

4.3.2.2 Dimensional examination.- Examination shall be made of the punch and die for defects in dimensions. Any dimension not within specification requirements shall be classified as a defect. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units.

4.3.2.3 Testing of the end item.- The sample unit shall be tested in accordance with [illegible]. The inspection level shall be S-1 with an AQL of 2.5, expressed in terms of defects per hundred units.

* 4.3.3 Examination of preparation for delivery.- Examination of preparation for delivery shall be in accordance with MIL-P-15424.

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* 4.4 Test.-

4.4.1 Hardness test.- The working surface of the punch and die shall be tested for hardness in accordance with ASTM E-18 to determine compliance with 3.4.1 and 3.4.2. Three readings shall be taken adjacent to the working surface of the punch and three readings shall be taken adjacent to the working surface of the die. The average of the readings shall be the hardness of the sample punch or die, providing each reading is within the specified range. Nonconformance to 3.4.1 or 3.4.2 constitutes failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packing, packaging and marking shall be in accordance with MIL-H-15424. Levels of preservation, packaging and packing shall be as specified (see 6.2).

6. NOTES

* 6.1 Intended use.- Punches and dies covered by this specification are intended for inserting grommets in duck, canvas, leather, rubber and similar materials. Type I punches and dies are intended for inserting eyelet grommets. Type II punches and dies are intended for inserting spur grommets.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Size of punch and die required (see 3.4).
(d) First article.- When a preproduction sample is required (see 3.1).
(e) Selection of applicable levels of preservation, packaging and packing required (see 5.1).

* 6.3 The requirements for the type III punches and dies were deleted from this revision since the items are no longer catalogued.

* 6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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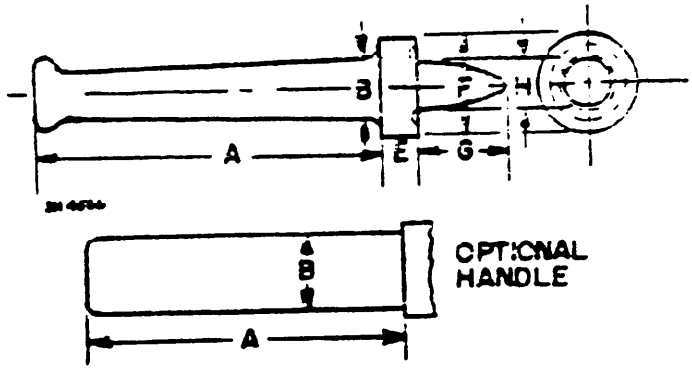
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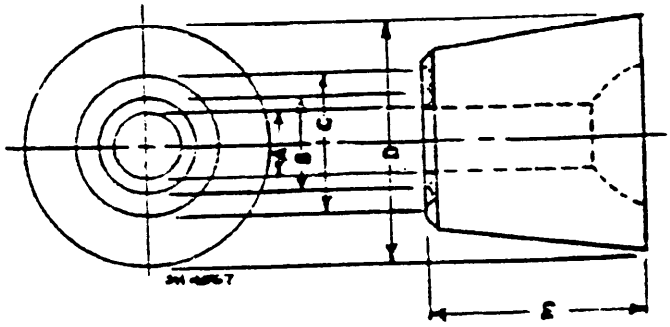
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Size Number	Handle		Punch end			
	Length	Diameter	Pins		Fillet	
			Thickness	Diameter	Length	Major diameter
	A ± 1/4	B ± 1/16	E ± 1/8	F ± .015	G ± 1/16	H — .005 + .001
	Inches	Inches	Inches	Inches	Inches	Inches
00	3-1/8	7/16	3/8	.515	9/16	.182
0	3-1/8	7/16	3/8	41/64	3/4	.254
1	3-5/16	9/16	1/2	51/64	13/16	.296
2	3-11/16	9/16	17/32	29/32	1	.375
3	3-13/16	5/8	9/16	1	1-1/16	.423
4	3-7/8	5/8	9/16	1-1/8	1-1/8	.500
5	4-3/16	25/32	5/8	1-5/16	1-5/16	.625
6	4-1/8	1	3/4	1-47/64	1-3/4	.800
8	4-11/16	1-1/8	11/16	1-25/32	1-11/16	.900
10	4-3/8	1-3/32	13/16	2-31/64	2	1.253
11	4-1/2	1-15/64	27/32	3	2-1/16	1.555
12	5	1-25/64	7/8	3-33/64	1-15/16	1.990

Figure 1. Type I, Eyelet Grommet Inserting Punch

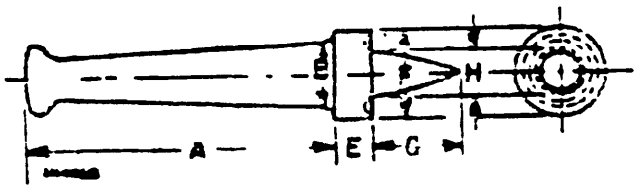
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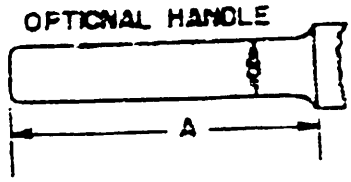


Size number	Die diameters				Height of die
	Center hole	Inside bead	Outside bead	Overall	
	A ± .005	B ± .015	C ± .010	D ± 1/4	E ± 1/4
	Inches	Inches	Inches	Inches	Inches
00	.187	———	.468	1-7/8	1-15/16
0	.254	.424	.553	1-7/8	1-15/16
1	.296	.624	.697	1-7/8	1-15/16
2	.390	.735	.822	1-7/8	1-15/16
3	.458	.874	.975	1-7/8	1-15/16
4	.500	.953	1.057	2-3/16	1-15/16
5	.525	1.125	1.228	2-3/16	1-15/16
6	.562	1.265	1.350	2-3/16	1-15/16
8	.687	1.406	1.537	2-15/16	2-1/8
10	1.281	1.750	2.375	3-5/16	2-3/8
11	1.578	2.125	2.812	3-7/8	2-1/4
15	2.046	2.562	3.312	4-3/16	2-3/16

Figure 2. Type I. Evelet Grommet Inserting Die

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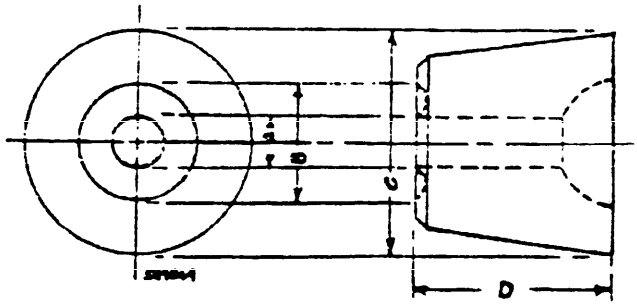




Size number	Handle		Punch end			
	Length	Diameter	Face		Pilot	
			Thickness	Diameter	Length	Major diameter
	A ± 1/4	B ± 1/16	E ± 1/8	F ± 1/64	G ± 1/16	H + .002 − .005
	Inches	Inches	Inches	Inches	Inches	Inches
0	3-1/4	3/8	7/32	22/32	3/4	.234
1	3-3/16	9/16	7/16	13/16	7/8	.375
2	3-11/16	9/16	1/2	15/16	1	.437
3	3-13/16	5/8	1/2	1-1/16	1-1/8	.468
4	3-7/8	5/8	1/2	1-1/4	1-1/8	.515
5	4-3/16	25/32	5/8	1-3/8	1-1/4	.625

Figure 3. Type II, Class 1, Integral Die, Spur Grommet Inserting Punch

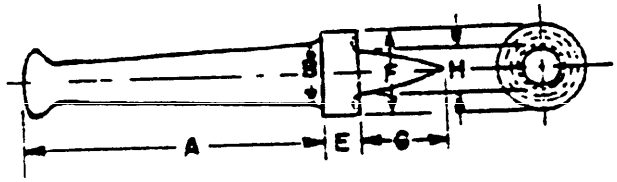
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Size number	Die diameters			Height of die
	Center hole	Bead	Overall	
	A +.005 −.000	B ± .015	C ± 1/4	D ± 1/4
	Inches	Inches	Inches	Inches
0	.296	.750	1-7/8	1-15/16
1	.375	.875	1-7/8	1-15/16
2	.437	.937	1-7/8	1-15/16
3	.458	1.062	1-7/8	1-15/16
4	.546	1.218	2-3/16	1-15/16
5	.625	1.375	2-3/16	1-15/16

Figure 4. Type II, Class 1, Integral Die, Spur Grommet Inserting Die

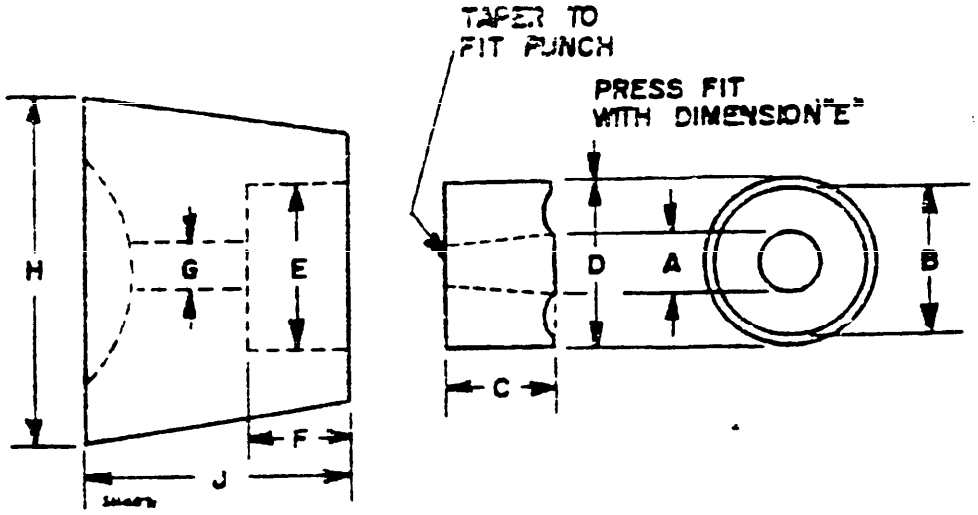
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Size number	Handle				Punch end			
	Length	Diameter			Face		Flat	
	A nominal	B nominal	C	D	Thickness E (min.)	Diameter F ±0.005	Length G ±0.010	Major diameter H +0.000 −0.005
	Inches	Inch			Inch	Inches	Inches	Inch
0	3-5/8	3/4			1/2	3/4	13/16	0.296
1	4-1/8	3/4			5/8	15/16	1	.406
2	4-1/8	13/16			9/16	1	1-1/16	.437
3	4-1/4	7/8			9/16	1-1/8	1-1/16	.468
4	4-1/4	1			1/2	1-1/4	1-1/16	.573
5	4-3/8	1			7/16	1-7/16	——	.625

Figure 5. Type II, Class 2, Inserting Die Spur Grommet Inserting Punch

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Size number	Die insert dimensions				Block dimensions				
	Center hole	Band	Length	O.D.	Insert cavity diameter	Insert cavity depth	Center hole	O.D.	Height
	A +0.005 −.000	B ±0.010	C ±1/32	D +0.001 −.000	E	F ±1/32	G ±1/32	H (min.)	J (min.)
	Inch	Inches	Inch	Inches	Inches	Inch	Inch	Inches	Inches
0	0.391	25/32	13/16	1	1	3/4	1/2	2	2
1	.411	25/32	13/16	1	1	3/4	1/2	2	2
2	.442	15/16	13/16	1	1	3/4	1/2	2	2
3	.473	1-1/16	13/16	1-1/4	1-1/4	3/4	1/2	2-3/8	2
4	.583	1-1/8	13/16	1-1/4	1-1/4	3/4	1/2	2-3/8	2
5	.630	1-3/8	13/16	1-3/8	1-3/8	3/4	1/2	2-3/8	2

Figure 6. Type II, Class 2, Inserted Die, Snur Grommet Inserting Die

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Punches and Dies, Grommet Inserting MIL-P-16208D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

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